

LITTLE BLUE BOOK NO.

Edited by E. Haldeman-Julius

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*** START OF THE PROJECT GUTENBERG EBOOK THE FACTS
ABOUT REJUVENATION ***

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The Facts About Rejuvenation

Morris Fishbein, M. D.

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THE FACTS ABOUT REJUVENATION

THE APOTHEOSIS OF MEDICINE

The medicine of the past was a medley of superstition, faith, theory and tradition, lightened by gleams of accurate observation and logical thought. The medicine of today is a science founded on actual observations of the manifestations of disease in man and in animals, in both living and dead bodies. The medicine of an early day was based largely on intricate hypotheses as to the causes of disease. There was no systematic search in the unknown in any way comparable to the steady, steadfast progress of medical science resulting from research during the last fifty years. The medicine of a previous epoch included only the physician who treated the individual patient. The medicine of today has changed this old profession, whose only function was the care of the individual sick man, into a dozen professions, which include not only the care of the individual patient but also the study of disease, its prevention, the attack on disease, and the care of the public health.

No doubt everyone can remember the family physician of some twenty or thirty years ago, who made his diagnosis largely by asking about the progress of the disease from the time the patient first noted symptoms, and by the use of such information as he could acquire with his unaided senses. It must not be thought, however, that the information which the physician acquires in this way is the less scientific because it depends merely on the use of the hand, the ear and the eye. More than one hundred years ago, two physicians made observations as to the way in which changes within the human body could be discovered by the use of these simple methods. Auenbrugger, in Austria, studied the sounds that occur when the human chest is thumped. Everyone knows that when you thump a cask that is half full of water the lower half will sound different from the empty half. Indeed, an experienced Negro can tell you the relative state of ripeness of a watermelon by a similar test. And Auenbrugger used this simple method to ascertain when the lung might be solidified by pneumonia, when the chest cavity might contain fluid or pus as a result of some infection, when the

heart was enlarged beyond the normal limits as a result, perhaps, of overstrain or disease. And when all of these things were carefully written down as a result of thousands of observations made by physicians all over the world, it became possible for those educated in this method to know, with a fair degree of accuracy, the condition that existed within the patient's chest.

A little later, in Paris, a physician named Laennec, while walking to the hospital, noted two boys at opposite ends of a log, one listening and the other scratching the log with a pin. He found that the boy listened because he could hear the sound transmitted through the log. The outcome of this simple observation was the discovery of the stethoscope, the device which the physician employs to determine the sounds that go on within the human body when the organs function. On listening to the heart in which one of the valves has given way, one may hear gurgling and murmurs which are the result of physical changes in the heart. On listening to the lungs, one may determine whether the little spaces are filled with air, as they ought to be when the patient breathes, or whether some disease has consolidated the lung and made breathing impossible. These observations are just as regular and certain as any mathematical formula; they are as scientific as the observations made by an engineer; they are as definite as the diagnosis of a broken valve or a carbon-choked cylinder in a motor car.

The textbooks in medicine of some fifty years ago were an unusual conglomeration of the accumulated facts of the centuries and of accurate observations made by careful physicians, mingled with queer ideas as to why things were as they seemed to be and unestablished rules based on the wrong interpretation of only partially understood phenomena. There was, for instance, certain knowledge that the bark of the cinchona tree, from which quinine comes, would stop the chills and fever of malaria. There was not, however, the fact, later shown by Laveran and by Rosa, that malaria is caused by an organism in the blood which is transmitted from one person to another by a mosquito. It had been observed that mercury would benefit syphilis, but no one knew that syphilis was caused by a spiral organism which may be transmitted from one person to another. It was known that certain drugs, such as digitalis, would slow the beat of the heart and strengthen it, but it was not known that it does this by action on the nerves leading to the heart.

The time, then, was ripe for a scientist like Pasteur, and particularly ripe for the attitude of skepticism with which he approached the established beliefs of his day. The credit for the great advance of the last fifty years in medicine must be given to several factors: First, a changing conception of the way in which disease is brought about in the human body, due to the establishment of what some people still call the “germ theory.” Germs are no more a theory than is poison ivy. The results of the action of germs within the human body are no more theoretical than is the eruption that follows contact with poison ivy in sensitive persons. The second great factor influencing medical knowledge was the application to it of facts and materials derived from such fundamental sciences as chemistry, physics, anatomy and physiology. And the third important development was the invention of mechanical accessories to aid the human senses in studying disease, both within and without the human body.

After Pasteur had shown that bacteria do exist and that they cause disease, Robert Koch, a German investigator and discoverer of the cause of tuberculosis, established certain laws for determining whether or not any single germ was actually responsible for producing any certain disease. It is unnecessary to mention specifically each of these laws; suffice to say that they involve the actual proof of the fact that the germ is in the body of a person dying from disease, and that the disease may be produced in animals by injecting them with the germ. It would be undesirable to mention in this article all of the diseases for which medical science now knows the bacterial cause; the list would include, however, such diseases as tuberculosis, syphilis, diphtheria, meningitis, typhoid fever, dysentery, pneumonia, lock-jaw, cholera, malaria, anthrax, scarlet fever and a vast number of less commonly known diseases. Hardly a year passes in which new germs are not isolated and in which obscure conditions are not brought to light. The recognition of this fact is of the greatest importance. If one accepts the definitely proved facts as to the relationship of bacteria to disease, one must at once discard all of those systems of healing which claim that all diseases are due to a single form of disturbance and are curable by a readjustment of that disturbance. There is not, for instance, the slightest scientific evidence that pressures on the nerves coming from the spine or on the blood vessels leading to the various organs are responsible for the many diseases from which the human body may suffer.

Now this does not mean that medical science is in possession of the facts regarding all diseases. Much remains to be learned. There are such diseases as measles, chicken pox, mumps, German measles, epidemic sleeping sickness, and even common colds and influenza, for which we have no germ that everyone will admit is the cause of the disease. But medical science, with its careful, systematic methods, has learned many facts regarding such diseases, and the application of these facts is important in their proper diagnosis and healing.

I mention particularly the ways in which medical science calls on mechanical aids in modern diagnosis. The modern physician utilizes mirrors and lights to observe directly the inner lining of the stomach, the bladder, the ear drum, the nose and the throat. You all know the story of the young physician who put the mirror into the woman's mouth and said: "Madam, you have no idea how far we can see with these things." She leaped from the chair and said, "I beg your pardon, doctor; I left home with a hole in the heel of my stocking." By means of the X-ray, the doctor is able to observe changes in solid organs within the body; and through the injection of air into the cavities, he is able, with the aid of the X-ray, to outline organs heretofore not visible and to determine the condition of tissues that cannot be touched. The laboratories have added information to be gained from studies of the blood and from all of the secretions and excretions of the body. Studies of the blood may show the type of infection from which the patient suffers; may show the manner in which he is responding to infection; may show whether he is anemic or has a sufficient number of blood cells to conduct properly the chemistry of his body. And these matters are not beyond the comprehension of the ordinary human being. They may be demonstrated to any intelligent person; in fact, in many laboratories these simple tests are regularly performed by the people who have been trained only in the tests and not in medical knowledge. The interpretation of the test, however, is always a matter for a well-informed physician. Some of the cultists or misguided persons who believe that all diseases are brought about by pressures on nerves or on ligaments, by vibrations, by the influence of the mind, or by any other narrow, restricted mechanism of disease causation, should begin to realize that their theories must be discarded in the face of such observations as these.

I pass over also the newer methods in the treatment of disease, involving the use of newly discovered antiseptics, of glands and gland substances, of vaccines and serums, and of unusual drugs. I pass over the beneficent effects of modern surgery and the newer anesthetics which have been developed and which enable the physician to perform surgical operations painlessly and safely, as contrasted with the conditions a century ago that made operating rooms bloody shambles. Even the brain, particularly unapproachable in a previous century, is now the subject of operations for the removal of tumors and the relief of abscesses.

One by one, great infectious diseases and plagues have been brought under control. Fever-infested swamps and the jungle of the tropics have been freed of such menaces as yellow fever, malaria, African trypanosomiasis, and other diseases transmitted to man by insects. Through education in the value of a periodic physical examination and through the teaching of personal hygiene, intelligent persons are being enabled to live more and more comfortably. The death rate of tuberculosis has been reduced by more than one-half; typhoid fever no longer menaces communities which are willing to pay the price for good milk supply, good water and proper sewage disposal; and inspection of fresh and prepared foods has practically abolished the menace of diseases formerly transmitted in this manner. Special attention given to the problems of infant mortality has reduced the death rate in the earlier years of life and an infant coming into the world today has a far greater expectancy of life than did one coming into the world fifty years ago. The child that might then have expected to live to somewhere between thirty-five and thirty-nine years of age may now confidently look forward, under all ordinary circumstances, to reaching the age of fifty or fifty-five years. What hope, then, of giving that child an opportunity to reach its allotted three score and ten?

Everyone knows from his own casual observations that a great many more people are living to a ripe old age nowadays than was the case formerly. Indeed, these observations are the exemplification of the statistics that have been cited as to life expectancy. The diseases that still remain to cut down the human being at an earlier age than that at which he might reasonably expect to depart this life through the actual wearing out of the machinery which is no longer able to renew itself include particularly the diseases of the heart, blood vessels and the kidney. Such conditions as

Bright's disease, a chronic inflammation of the kidney; as hypertension, exemplified through high blood pressure and the symptoms that accompany it; as arteriosclerosis, or hardening of the arteries; as angina pectoris and inflammations of the lining of the heart and of its valves—all such conditions constitute a problem which the medical profession must meet and which it apparently cannot meet properly without the complete education and cooperation of the public. Here are diseases, the prevention of which seems to lie in the prevention of infections of the throat in childhood, in the training of the child and man in the proper hygiene of the nose, throat and intestinal tract; in the teaching to the adolescent and equally to the adult of the desirability of a quiet, well-ordered existence with a proper amount of food, exercise and rest; and, not the least by any means, in letting people know that there is such a thing as mental hygiene, including relief from worry or mental strain. And then there is pneumonia, a disease likely to affect the older man much more severely than the young one, called, indeed, “the old man's friend” because of the quick death that it produces in the senile; and the “captain of the men of death” because of its severely fatal character. The complete conquering of this infection seems somehow to be resting on the threshold of the future.

It is the belief of several competent statisticians who have given careful consideration to the matter, that when these diseases are brought under control, the life expectancy will be moved forward at least another five years, but probably not a great deal more. The trend of modern life seems to be toward speeding up, toward greater mental strain and almost ceaseless activity. If, coincident with such speeding, medical science is still able to promise another five years of life expectancy to the phenomenal number of years that have thus far been added, it will have accomplished a marvel beyond the dreams of the medical scientists of fifty years ago.

THE FACTS ABOUT REJUVENATION

There is no fool like an old fool—particularly in matters of rejuvenation. For the senile, tottering old man, the world has had only the pity that it confers on a Faust, who bartered his soul for a few years of youthfulness; the ridicule that it darts upon a Don Quixote; the mild amusement with which it listens to the tale of Ponce de Leon; or the savagery with which it

attacks the decrepit prey of physical lust, inspired by inflamed tissues, who slakes his appetite with the inveiglement of girls. Psychologists observe these as the expression of the law of self-preservation, the reaction of the living toward approaching death, the feeble call on unresponsive nerves and muscles for power that they cannot give.

Today the public is told again and again that the secret of restoring youthful vigor to worn out tissues has been solved in the laboratories of the Austrian Steinach and of the Frenchman Voronoff. Alas! medical scientists, who like to be shown before they are convinced, mistrust the evidence. They grant readily that the senescent graybeard with the will-to-believe may be instilled with young ideas, but they have yet to admit that any surgical operation, any transplantation of glands, any decoction of plant, organ or mineral remedies will confer on the withered form the machinery to put those ideas into action. Of the mechanical contrivances, the glandular extracts, and the other forms of hocus-pocus sold with claims for their ability to rejuvenate more will be said later. Here the disciples of the scalpel who are reaping something of a harvest in the field of surgical rejuvenation will receive discriminating attention.

Man's search for the elixir of eternal youth was never a scientific one previous to the period of Brown-Sequard. Long before that famous French physiologist, however, scientists had been studying the nature and the functions of the glands. It had been shown that the ability to have children was co-ordinate with certain changes in the body of the growing child. The voice of the boy deepens, causing great embarrassment when a fine soprano statement suddenly changes to a basso growl. The first inklings of mustache and beard send him to his mirror for hours of painstaking scrutiny. He begins to take a more serious interest in the female of the species. And it was taken for granted that the development of these characters was the result of a special or internal secretion poured into the blood by the glands. In science, however, it is not well to take anything for granted.

In 1889, the famous French physiologist thought that he had discovered the potent substance in extracts of the glands. One need not be a Freudian to realize the suggestion conveyed by the terms vim, vigor, virility or vitality. It was not strange, therefore, that attention should be turned to the peculiarly male organs in the search for the important substance. Indeed, the most primitive savages of cannibalistic nature were wont themselves to

brew essences of the organs of the enemies whom they slew in battle, believing that the eating of the tissues was sufficient to confer upon them the prowess of their vanquished foes. Nor is it surprising that Brown-Sequard, having inoculated himself with extracts of glands, developed a sort of similitude of youthfulness. He reported that he could now climb a flight of steps with greater rapidity and ease than before his inoculation. Nevertheless, at the appointed time, the body of Brown-Sequard went the way of all flesh. And the skeptical scientists who repeated his experiments, checking them with numerous cases and controlling them with injections of plain water instead of the extracts, shook their heads in dismay. The great scientist—for Brown-Sequard was all of that—had yielded science to enthusiasm and his conclusions could not be sustained.

THE THEORY OF STEINACH

The conception on which the Steinach operation for rejuvenation is based is not easy to comprehend. About 1903, two French biologists claimed that the secretion responsible for maleness was developed by a certain part of the male tissues. They asserted that this portion developed particularly if the tubes were tied off. And Steinach claimed that he had confirmed their experiments and that the performance of this operation on old animals, particularly rats, resulted in rejuvenation.

Now the finding of a substance or a system of rejuvenation is much like the finding of gold. Whether it is there or not, all of the unsuccessful, romantic, and adventurous experimenters rush in on the trail, and all of the aged and worn out capitalists come in as soon as convenient to take advantage of the discovery. The cry of “gold, gold” was taken up quickly by a number of young and enthusiastic investigators, who followed Steinach. Some famous actors, physicians and financiers who saw the waning of their power and note and of their ability to enjoy to the utmost the lives that had given them so much became subjects of the experiment. Where there are actors, authors and financiers, there is also always, in these modern times, good publicity. And where there is publicity not too careful as to the facts, there are soon more applicants for operations. This, in medical science, is known as a vicious circle.

The careful experimenters who have followed the work of Steinach, checking all his results carefully on experimental animals, have demolished his claims bit by bit. Oslund of the Vanderbilt Medical School showed that the Steinach adherents had been deceived because of the type of experimental animal used. Since the theory of the rejuvenescence is based on an over-development of cells that does not actually occur, it can be taken for granted that cutting or tying of the tubes with the idea of producing such over-development cannot cause rejuvenation. And the work of Oslund has been confirmed by many other investigators of established standing.

Moreover, the belief that the male glands alone are responsible for the development of male characteristics has been completely demolished. The complete male features of mind and body, including the external organs peculiar to the male, have been found in persons in whom the so-called male glands were absent from birth, as proved by complete postmortem examination. Furthermore, all of the features of the male sex have been found in persons in whom the male glands were absent from birth, but who had within the body a complete set of female glandular organs. There is a great deal of much more technical evidence to support the view that the male glands are not the only tissues concerned with establishing maleness or femaleness of any individual. Perhaps the best of it developed in experiments on hens and roosters, revealing hens possessing the comb and wattles and even the feathers of the rooster, yet living a hen's life, laying eggs and hatching them out! But what, you ask, has all of this to do with Steinach's theory of rejuvenation?

The elderly gentleman who prefers blondes is hardly likely to be deterred by the details of technical experiments on mice, rabbits, chickens, or dogs. For him, therefore, one presents the facts regarding man. For many years the operation now called the Steinach rejuvenation operation was performed on old men who suffered with a disease quite common in old men. But in not one of the hundreds of cases in which the tubes were cut or tied as in the Steinach operation did any of the meticulous surgeons mention any restoration of youthful vigor or anything resembling rejuvenation. This fact has been called to the attention of Steinach and of all his followers again and again in reputable medical publications, but an explanation has never been offered.

To the proof of the scientific laboratory investigators that the Steinach method is founded on a fallacy, the Steinach adherents and the surgeons who perform the Steinach operation reply with the sort of cynical shrug of the shoulders that such practical men use in answering laboratory evidence. They point with none too reluctant pride to their records of cases operated on and to the letters of testimony. Here is as simple a gesture as might be made by a moron, but hardly the sort of thing one expects from a scientist. The same testimonials—in fact much stronger ones—can be found for all of the nostrums and contrivances that have invaded this fertile field.

Opposed to the enthusiastic statement of Adolph Lorenz is that of Professor M. Zeissl of Vienna, that the only change he noted after his operation was less interest in sexual matters than previously. Opposed to the quoted interviews of Dr. Harry Benjamin of New York concerning the one hundred cases on which he operated with a claimed considerable percentage of success are the records hidden away in medical literature of thousands and thousands of men on whom such operations were done without any noticeable response.

There are, indeed, some records of senile men who put upon their degenerated tissues far more of stress than they could tolerate. Here the illusion of vigor produced an earlier death than might reasonably have occurred without the operation and in the absence of the illusion.

The prime exponent of the Steinachian miracle in this land that so readily accepts overnight miracles in medicine seems to be Dr. Harry Benjamin. In his earlier publications one finds no mention of any possible mental defects associated with the surgical procedure. But quite recently, regardless of the fact that he is without status in the field of psychology and psychiatry, he begins to mutter somewhat vaguely concerning the importance of coincident psycho-analysis. And in the publicity concerning his efforts appearing within the past few months appears what is, for so shrewd a man, a most incautious admission: "The experts declare that the value of the Steinach method has been much reduced by ignorance and by the opposition of doctors who have mistaken ideas regarding the operation." In other words, the patient must believe thoroughly that the operation will give him vim, vigor and vitality, or he doesn't get it. For many years, physicians who have specialized in the diseases peculiar to men have emphasized again and again the great part played by the mind in controlling the body

functions. The records of the surgeons who do the Steinach operations are without scientific controls. To what extent have they checked their work by cases on whom the operation was not done, but who otherwise were put through the same procedure? Obviously such controlling of cases would be a most elementary and simple procedure, but records of such control in their publications are sadly lacking.

Indeed, these exponents of surgical art would seem to be providing for the future in their claim that the operation may be repeated with good effect and a second revival of powers bestowed on the waning patient. Here is a biological revolution that bewilders the imagination! The young men, handicapped in competition because of lack of experience and of the world's goods, will be compelled to struggle against their rejuvenated fathers and grandfathers, while the latter, by repeated operations, maintain a permanent lead. It's a sad, sad outlook; but fortunately it isn't true.

THE VORONOFF CONCEPTION

The contention of Voronoff, like that of Steinach, has to do with the use of gland material for purposes of rejuvenation. He does not, however, depend on any change in the tissues brought about by tying off tubes. He actually transplants the entire glands, and since the human material is not easily and generally available, attempts to supply the deficiency with glands derived from the anthropoid apes. In this instance also a vast amount of experimental evidence from the past and from other workers of the present is available to disprove the contention. True, there are instances in which the glands have been absent from birth or lost through accident and in which the transplantation of such material results favorably, but the records again are not such as will convince any scientific reader of the general merit of the operation. In the first place there is not in any sense of the word actual prolongation of life, since none of those on whom the transplantings have been done seem to live beyond the normal period. In the second place, one reads such conclusions as the following: "The pessimistic attitude of the patient and constant brooding over his inability have marred the results of the treatment." Here again the part played by the will-to-believe seems paramount.

When all of the evidence is assembled and considered *en masse*, it becomes apparent that there is not as yet any actual proof that rejuvenation has been accomplished in a single individual, or any basis for the belief that it ever will be accomplished.

REJUVENATION: THE MECHANICAL AND GLANDULAR METHODS

Rejuvenation is, in these days of young old-men and old young-men, a word to conjure with. The astute purveyors of mechanical devices, pills, lotions and systems of physical culture sold by the mail-order plan have been quick to realize the wonderful “come-on” possibilities inherent in the word. The result has been the interminable repetition of the claim for renewed youth achieved through hundreds of substances and appliances. The post office department itself is hardly able to keep pace with the new developments in the field and its fraud orders are issued usually after the promoters have reaped a fine harvest.

MECHANICAL REJUVENATION

The senescent man or the youthful individual who finds himself suddenly lacking in sexual vigor is a ready prey for the exploiters of mechanical devices, urged because of their ability to encourage a physical development that seems to be lacking. The average man in this land of unenlightenment regarding the physical constitution has but slight conception of what constitutes the normal in these affairs, either as to physical growth or functional ability. Indeed, the more he meditates upon the matter, the more he is likely to be convinced that he himself is sadly lacking in these particulars, the psychology of this view being dependent upon a will to achieve rather than on actual knowledge. Moreover, the cleverly worded literature—and what a sad use of the word this is—of the salesman of various devices is calculated to intensify and emphasize this view. All of the advertising matter is a paean of praise of sexual athleticism, and a weeping and wailing over deficient performance, inspired to make still more mournful the sexually despondent male.

On December 5, 1925, the post office department issued a fraud order against the manufacturers of a device known as the "Perfection Developer," one Walter H. Hartman of Columbus, Ohio, selling under the firm name of "Hart & Co." The simple device was merely a cylinder of glass connected with a pump and designed to induce a vacuum. The effects persisted only so long as the device was in actual use, and if it had any permanent effects whatever, they could only be for harm.

Another device of the same type was sold by one M. Von Schwartz and William Billings of Ithaca, New York, under the name of the "Burt Vacuum Tube." This firm was barred from using the mails September 1, 1925.

These devices are typical of most of the mechanical appliances offered by mail for men lacking sexual vigor. Obviously, they can be of no permanent service. Moreover, the psychology of their use is such as to discourage any real possibility of improvement through rest, diet and psychotherapy which long experience has shown are of value.

GLANDULAR REJUVENATORS

Public interest in any new medical development is at once capitalized by the promoters of nostrums and fallacies. It is not surprising, therefore, that increasing interest and knowledge of the glands of internal secretion should have aroused in the minds of some promoters the possibility of unusual and unwarranted gains.

In March, 1921, the "Youth Gland Chemical Laboratories" was incorporated in Illinois; in February, 1922, the name of the corporation was changed to the "Druesen-Kraft Chemical Laboratories." Later, it became known as the "Lewis Laboratories." It was not, however, until March 19, 1925, that the United States Government closed the mails to this concern. In the meantime, it reaped a rich harvest from the unwary. The advertising literature of this concern was of a style so striking as to have merited some better use. A full-page announcement in the newspapers was headed, in two-inch, black type, "YOUR GLANDS WEAR OUT"; in the center of the page, he-men and she-women danced the one-step in close embrace. And here are the phrases that brought the replies:

—It is based entirely on the principle of Feeding Actual Gland Substance Direct to the Glands, thereby renewing and rejuvenating them.

—This method of giving new life to the glands is advocated and endorsed by the leading students of gland therapy throughout the world—including Dr. Arnold Lorand, who is generally conceded to be the greatest living authority on this subject.

—The actual Method of Treatment used by us is the result of exhaustive experiments covering several thousand cases, during the past two years.

—The “Lewis” Treatment Is Practically Never Failing.

—If You Could Prevent the Wear and Tear on Your Glands Caused by Sickness, Age, Disease, etc., You Would Look and Feel as Young at 70 as at 25. Science However Has Solved the Secrets of the Glands and Now for the First Time Shows You the True Way to KEEP or REGAIN Your Vigor by Feeding and Replenishing the Most Important Glands!

—Build Up Your Glands and You Build Up Your Strength and Endurance.

The evidence is clear that the firm did exceedingly well from a financial point of view. The original price of the treatment was \$10.00, which, on occasion, could be reduced to \$7.50, and, if the victim still failed to succumb, to a special offer of \$2.95. The court records in the case showed that the firm had spent at least \$300,000.00 in advertising, and that it had a gross annual income of between \$250,000.00 and \$300,000.00. Nevertheless, scientific evidence has shown that extracts or other preparations of the sex glands are without power when taken by mouth. They not only fail to produce rejuvenation of the entire body, but even to stimulate to any extent the particular portions of the body in which the applicant for rejuvenation seems to be especially interested.

Another concern, with headquarters in Denver, issued a booklet entitled “The Secret of Staying Young.” Here were the testimonials of elderly men, 82, 83 and 84 years of age, who announced that they had lasting benefits from the treatments offered. These testimonials are just as vociferous and emphatic as those of aspirants for youth operated on by the disciples of Steinach. The treatments consisted of nothing more than dried animal glands, taken by mouth, which, as has been said, could not possibly have

the effects claimed for them. The concern exploiting these desiccated glands, the “Vital-O-Gland Company,” not content with depending on the mental responses of those who took its preparations by mouth, sold at the same time the usual vacuum developer, a glass tube attached to a bicycle pump. When the Government investigated this concern, it found twenty-two girls and five men occupied in sending out the salacious literature. The evidence revealed that the gross income of the concern for 1923 was \$176,406.82. Rejuvenation pays!

The Vital-O-Gland Company and the Lewis Laboratories have been barred by the Government from the use of mails, but there still remain many concerns offering, both to physicians and to the public, glandular preparations for rejuvenation or for sexual stimulation. There will always be senescent, somewhat lewd and sad old men to waste the funds of their declining years on such powerless pills. The memories of youth become more and more resplendent with the passing years.

The Glandine Laboratories of Chicago and Los Angeles issues a circular with the question, “Must We Grow Old?” and with the answer, “Science Says, No.” The treatments consist of extracts of the sex glands to be taken by mouth. The Glandex Company of New York advertised in the public press a combination of gland extracts with iron. The International Research Laboratories of Chicago advertised “Baker’s Glandol,” with a salacious pamphlet emphasizing the rejuvenation of man, and with such headings as “The Most Interesting Thing Is Love. Don’t Waste Life. Luck from Boldness and Suppressed Desires.” Where is the man who could resist the plea?

The Puritan Laboratories of Nashville, Tennessee, issues “Glandtone,” offered with the claim that it will restore youthful vigor to those passing with age. The Walton Chemical Company of Chicago emphasizes the glands and says that a combination of the sex glands, the thyroid, prostate, pituitary and adrenal, hermetically sealed, will defer old age and renew vitality. What of the impudence and psychological cleverness of its warning:

WARNING!

The country is being flooded with literature from so-called “Laboratories,” offering to restore Sex STRENGTH in the forms of

unsealed “Gland Tablets” or “Liquids.”

Such preparations may contain some gland substance, but unless the ingredients are SEALED to preserve their strength, the potency may be entirely gone in a few days.

The Walton Treatment INSURES the STRENGTH of the INGREDIENTS and thousands have found this method SUCCESSFUL even after years and years of previous failure with other methods of treatment.

In fact, fully three out of four who write of their remarkable results with this new method, say they had taken many other Treatments or methods without relief.

Bear in mind that there is not an iota of evidence to show that any preparation of male sex glands, singly or in combination, has ever been shown to have the slightest effect on the human body when given by mouth. It was the hope and the belief of Brown-Sequard that he had found such a substance. Even he thought that the taking of sex glands brought about in him a sort of rejuvenation. Yet in the more than fifty years since his passing, and since his claims were disproved, the public has not learned the truth.

RADIUM AND LIGHT

With the coming of the newer discoveries in medicine as to the effects on the human body of radium, the X-ray and ultra violet light, the agile-minded exploiters of the public's interest in keeping young were again quick to respond with the preparation and sale of such apparatus and material for purposes of rejuvenation. Of course, rejuvenation is not a physical state that can be easily determined accurately. The old man with declining powers is ready to welcome the slightest sign of increasing ability in physical work or in sexual activity. He forgets to reason that the rest and the mental stimulation associated with any new method of treatment are likely to bring about the illusion of strength.

The manufacturer of a cabinet lined with incandescent lamps, for example, is not content to claim for it the simple uses of a sweat bath, or of heat produced by incandescent lamps. Ah! No! This simple cabinet

becomes as by the wave of a magician's want, an "Inductive Metabolizing Method." And the manufacturer says:

The —— Inductive Metabolizing method gives to the world one of the greatest, if not the greatest, means of rejuvenation of the human organism known to medicine, but the part that is most interesting about this modality is that of its synergetic action with all the recognized methods of rejuvenation now in use—such as gland therapy, radioactive drinking water, baths, etc., as the results obtained by their use, when accompanied by the Inductive Metabolizing treatments, are increased tenfold.

One can almost picture him rubbing his hands in glee as he murmurs: "Rejuvenation: That's the word that gets 'em!"

The makers of apparatus containing radium or the blenders of waters which have been submitted to the rays of this wondrous element urge also its potency for rejuvenation. They point knowingly to the radioactive springs of Germany and Switzerland and cite the records of the old men who have visited those springs and returned home younger in body, if not in years. But they too neglect the effect of weeks of rest and freedom from care, of good diet and salubrious surroundings, and possibly—in fact, probably—of enforced inactivity for those portions of the anatomy whose physical functioning persists in remaining the center of interest when rejuvenation is discussed.

"THE NEW SCIENCE OF RADIENDOCRINOLOGY"

If radiation rejuvenates—although of course it doesn't—and if glands rejuvenate—and it has been shown that they do not—then, say the manufacturers of the Radiendocrinator, the combination will get the result. Merely ray the glands with the Radiendocrinator—price \$150.00—and you are there! The booklets are a hodgepodge of exaggerations, fallacies, and confusions regarding chemistry, physics, biology, glands, radium, and whatnot. But the convincing document is a blue-colored bonded guarantee of satisfaction or money refunded. Couched in language that seems to make the chance of financial loss to the unwary impossible, the guarantee is nevertheless a snare and a delusion. Launched originally by Dr. Herman H. Rubin, New York, as a device to be worn at night over the glands and

selling at \$1,000.00, the apparatus is now a reflector on a stand and can be purchased for \$150.00 But for pure romance the literature is at least ten times as fanciful.

THE HOPE FOR THE FUTURE

Even the Greek philosophers murmured that physical powers in matters of sex did not always parallel brain capacity. Strange that the sex instinct should be so deeply rooted and so prominent in life that two thousand years of experience has failed to convince men of the truthfulness of that statement.

Modern science has shown that the early detection of signs of disease by physical examination, and the establishment of a proper regime of life, including adequate rest, diet, exercise, simple personal hygiene, and freedom from worry will greatly extend the life of the average man. He may quite reasonably hope to live to seventy years and if he is at all careful considerably beyond that age. It behooves him then to grow old gracefully, remembering that much of the greatest work of this world done in art, letters, invention, finance and statesmanship has been done by men well beyond sixty years of age. Their minds were perhaps little given to the purely pleasurable functions of the bodies which constituted the abodes of their restless spirits.

THE ANTIVIVISECTIONIST AND ANIMAL EXPERIMENTATION

George Jean Nathan, in that immortal document *Pistols for Two* in which he and Monsieur Mencken, using the cognomen Maj. Owen Hatteras, gave an unsuspecting world an insight into their personal characters and characteristics, tells us which of his aphorisms delights him most. Of all the soul-searching mots perpetrated by this astute coiner of phrases, the one he selected is likewise the one that gives me most joy. "An antivivisectionist," said Dr. Nathan, "is a woman who strains at a guinea pig and swallows a baby." There it is in a nutshell! A Freudian might claim that the term "nutshell" was prominent in my mind because I was discussing the antivivisectionist.

At a period when the whole world begins to turn to science as the real goal of mankind; when intelligent human beings begin to discard pseudo-sentiment for fact, the followers of what is essentially merely an illogical, fanatical cult continue to oppose progress if it is to involve in any way what they conceive to be abuse of the lower animals for purposes of study. This opposition seems to rest invariably on a lack of actual knowledge of what animal experimentation has accomplished for mankind, of what it has contributed to the life and comfort of the animal, of the extent to which the animal may suffer in the cause of experimentation, and of the very rules which the scientists themselves have elaborated to safeguard their work with animals.

It is impossible in the scope of a brief article even to enumerate all that has been learned by animal experimentation. Without the aid of this method Pasteur could not have founded the science of bacteriology; such diseases as hydrophobia, tuberculosis, yellow fever, plague, scarlet fever, diphtheria and diabetes would not have passed under the control of scientific medicine but would have continued to take their immense toll of life and to cause immeasurable economic loss through illness. Indeed, to calculate the sums saved and earned through preventing illness, through opening up countries constantly menaced by disease, through the building of the Panama Canal, and through the saving of workers in industry, would produce a figure so vast as to be almost incredible.

Anyone who has seen a child, succumbing to the gradual encroachment of the diphtheric membrane in its throat, suddenly respond to the marvelous effects of diphtheria antitoxin will oppose to the utmost any attempt to deprive that child of the remedy. In preparing antitoxin the horse is required for the production of the serum and the guinea pig for standardization. I have seen the horses used for such service. Their paths are spread in pleasant places. They toil not at all, they are kept clean, they frolic in the open air except when the weather may be inclement. They are well fed and given the best of attention. I have seen the serum withdrawn and seen the horse make no more visible sign of protest than is made by the average man when he sticks his finger on a pin in the back of his wife's dress. I have seen guinea pigs by the thousands utilized for this work. If I am any judge of guinea pig emotions they do not suffer unduly in the process. I have

never seen a guinea pig suffer as much as a hysterical antivivisectionist suffers at a dog and pony show or a circus.

Before the discovery of the serum for epidemic meningitis from fifty to seventy-five percent of all who were afflicted died. When the serum is given the mortality is below twenty-five percent of those affected, and this says nothing of the saving in illness and in permanently disabling after effects. In the investigation which led to this discovery rabbits, guinea pigs, horses and monkeys were employed. In a war, nations sacrifice the lives of hundreds of thousands of their finest young men; the people are told that the sacrifice is made so that women and children may have a safe place in which to live. Shall the lives of rabbits and guinea pigs be weighed against the lives of the same children threatened by far greater dangers than those of war: the dangers of infectious disease? Puerperal infection which killed the woman in childbirth was controlled through animal experimentation. What about saving women from that menace?

It is not alone the so-called biological remedies such as antitoxins, vaccines and serums that are dependent on animal experimentation. Most of the potent drug remedies used today for the alleviation and cure of disease must be tested and standardized by the use of animals. Most of our important synthetic drugs were first evolved only by the use of animals. The anesthetics such as ether, chloroform, ethylene and nitrous oxide gas have the same effects on animals that they have on man. Was it not right that they should be tested first on animals? Such drugs as digitalis, which makes the lives of many persons suffering from heart disease endurable, have to be standardized by animal tests. Shall the poisonous doses of potent drugs be learned by tests made on the white rat or the guinea pig or on man; or worse still, shall we permit men to die or suffer mutilation to spare the feelings of the white mouse? Isn't it after all a question of sparing the hyperesthetic sensibilities of some idle woman rather than the duller sensibilities of some lower animal?

One by one the infectious diseases that attack man are being brought under control. There still remain those incurable diseases such as cancer, those conditions such as Bright's disease, high blood pressure, heart disease and diabetes, which may be alleviated but have not been completely conquered. Shall the methods that have brought success in the control of some diseases be discarded at the whim of misguided and unreasoning

followers of this cult opposed to human progress? Of course, animal experimentation never will be discarded. After all, the progress of scientific medicine is a powerful movement sweeping on and on with ever increasing impetus.

ANIMAL EXPERIMENTATION BENEFITS THE ANIMAL

Strangely enough the lower animals have benefited as much as has man himself through the progress made by animal experimentation. Hog cholera serum, the tuberculin test for cattle, the control of hoof and mouth disease, puerperal sepsis in cattle, hydrophobia, fowl plague, and the many worm diseases that afflict animals depend for their control on the same type of experimentation that yielded results for the diseases of man.

There is another side of the question that has its humorous aspects. It is necessary to kill hundreds of thousands of stray dogs and cats in our large cities to keep them from overrunning the human inhabitants. It has been estimated that a single pair of rats, if permitted to breed unchecked, might produce within three years a number of progeny running into eighteen or twenty-four figures. Everyone knows about the prolificity of guinea pigs and rabbits which multiply inordinately. Who shall say that these uses of science are not as kind a fate, as satisfactory an existence, as is the hunting down of the surplus by the hunter, the trapper, the poisoner, or the poundmaster? As we have shown, the horse used for the production of serum has a far happier life than the drawer of burdens in the ownership of some unthinking careless human being.

RULES CONTROLLING SCIENTIFIC EXPERIMENTATION

Medical scientists have not waited for government or other supervision to establish control over animal experimentation that will make it as nearly perfect as possible in preventing unnecessary pain and in providing animals with the best of care. A committee of the American Medical Association regularly functions for no other purpose. Under the control of this committee each laboratory binds itself to observe the rules laid down and to post those rules regularly in each department. The rules require that animals be held at least as long as they are held at the city pound; that they receive

every consideration for their bodily comfort; that no operations be made, except with the sanction of the director of the laboratory; that animals be anesthetized and rendered incapable of receiving pain in all operations, except in those in which anesthesia would defeat the object of the experiment, and finally that animals be killed painlessly at the conclusion of the experiment. These rules are most rigidly enforced and laboratories throughout the country are open to inspection by anyone interested from a scientific point of view, or from the point of view of control of this work.

If it be true, as has been said again and again, that science is the hope of the future for the progress of humanity, those who obstruct this progress by needless and unwarranted follies should be considered as subjects for mental investigation, or else as misguided sentimentalists whom one condones, but whom one does not take too seriously.

Transcriber's Note:

Inconsistencies in hyphenation have been standardized.

Minor punctuation errors have been changed without notice.

Spelling was retained as in the original except for the following changes:

Page [1](#): “D. Haldeman-Julius” to “E. Haldeman-Julius”

Page [5](#): “such as digitalis, woud” to “such as digitalis, would”

Page [10](#): “Such conditions as Brights” to “Such conditions as Bright’s”

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